Integrated Disaster Detection System for Flood and Fire

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***Abstract*—This research aims to develop a detection and alert system for two disasters, namely fire and flood, integrating Bluetooth communication and local alert mechanisms. This system is relevant to disaster management and early warning, hence bringing effective actions to reduce possible damage and loss of life. Unlike most available consumer solutions that detect either a flood or a fire, this project integrates both hazards into one system. Multiple sensors are integrated which are Ultrasonic sensor, Smoke sensors(MQ-2, MQ-4), and rain sensor, hence these will ensure a real-time response to disasters. Alert messages are conveyed through a Bluetooth module (HC-05) that will transmit the messages to an app on a mobile device. Additionally, LEDs and a buzzer with different sound rhythms are used to differentiate between the type of disaster detected which provides immediate localized feedback. This can be used in urban and rural setups to ensure security in areas prone to natural disasters or fires. A viable, cost-effective solution for providing life and property security is offered.**

***Index Terms*—Arduino UNO, Flood, Fire, Sensors, Bluetooth module.**

I. Introduction

Many calamities like floods, fires and gas leaks can devastate urban and rural areas. Floods can occur due the overflow of the water because of heavy rains, dam breaks and fires such as wildfires are uncontrolled blazes that spread quickly. There is lack of early detection of such disasters due to less infrastructure and less advanced systems usually in rural areas and even though the urban areas have proper resources have a greater challenge due to increased population and more property at risk. There are many cases where both the floods and fire happen simultaneously like short circuits during heavy rains which poses complex challenges for emergency response teams.

The aim of this system is to tackle the current system that lacks real-time and efficient communication in detecting two disasters which are flood and fire. Traditional system delays in response worsen the disaster severity due to the loss of human life and property. Traditional systems are also quite expensive and their installations in rural or developing areas become a challenging task due to infrastructure problems. This research focuses on developing an emergency response system that can detect disasters in real time and send localized alert messages using Bluetooth and simple alert mechanisms where the internet connectivity is unstable. The system aims to reduce the cost of installation and operation.

This proposed system integrates multiple sensors like Ultrasonic sensor for detecting the distance or range of the water level and rain sensor to detect rainfall, these both are used for detecting floods, and sensors like methane gas sensor to detect fire via gas leakage and smoke sensor to detect smoke and fire. Integration of many sensors minimizes interference and gives accurate readings. The alerts will be sent via a Bluetooth module (HC-05) which passes notifications to a Bluetooth Terminal app installed in a mobile device. An immediate feedback is provided through LEDs and different sound patterns given by a buzzer for various disasters. Therefore, the system is fully autonomous, which limits the time between the detection of a disaster and its action. it is cost-effective and scalable to both rural and urban premises.

Some approach addresses limitations in current fire and gas detection systems [1], while another focuses on flood prevention through DCOTech, a water-level management system that helps prevent flooding in residential areas [2]. The gaps addressed by our proposed work include combining the detection of fire and flood in a single framework. Integrating multiple sensors improves the monitoring capabilities and cross-verification and aims to improve the accuracy and reliability of hazard detection. The approach used is cost-effective all the components are affordable for implementation in both rural and urban areas, the system promotes low cost wider adoption, and scalability. The integration of Bluetooth communication, LEDs, and a buzzer ensures immediate awareness of disasters.

The contributions of this research includes:

- Detection of two disasters (flood and fire).

- Bluetooth module integration along with simple alert mechanisms.
- Multiple sensors integration for flood and fire disaster detection.

The rest of the paper is organized as follows: Section II is about the reviews of previous relevant work done in this field and how the proposed system is different. Section III states the system model of the proposed idea and Section IV is about implementation details. Section V is the result of the proposed experiments. Section VI is the conclusion i.e. how the objectives are achieved by this proposed method and what work can be done in the future to avoid the limitations of the present work.

II. Related Works

Development in fire detection technologies are described with much emphasis on vision and sensor-based systems amounts to improvement that must be dealt with [3]. Vision methods have to face challenges along the line of an urgent detection, on account of the various features of smoke and visual noise, while sensor technology can promote more economical installations coupled with increased reliability to help virtually eliminate false alarms.

The Early Flood Detection System developed in this work has been designed to send timely warnings to flood-prone areas, using an Android application along with ZigBee technology [4]. Pressure sensors installed in jack wells measure water levels. Wireless transmission of the data is done by a Raspberry Pi-based server station. By comparing current water levels with threshold levels derived from historical flood data, users are notified through an SMS and user-friendly Android application. The design permits independent work without the internet, assuring low power consumption, greater scalability, and universal applicability, thus offering an affordable and practical alternative for disaster preparedness in the resource-poor areas.

Another research shows that the inexpensive method of sending SMS alerts through the Arduino with GSM module allows monitoring gas and temperature levels, unlike the limited video monitoring method, automation through the use of Zigbee is greatly dependent on technical expertise [5]. In other words, UAV can provide awareness of fires to firefighters, though they may also be quite expensive. Recent breakthroughs instead stress using multiple sensors for near-instant warning and low-cost solutions to increase safety and efficiency in response time.

This research shows that different flooding monitoring techniques have their advantages and disadvantages [6]. While ultrasonic sensors impact distance of the water level measurement, they can be hampered in terms of accuracy by fog and rain. Gsm provides information as an SMS on a real-time basis yet may delay it owing to network issues. Arduino microcontrollers are cheap but possess very limited capabilities in signal processing. Most recent advances incorporate multiple sensors with Gsm to allow improved real-time monitoring and awareness.

The work discusses about a Wireless Flood Monitoring System that comprises an ultrasonic sensor and Arduino microcontroller used to monitor water levels, categorizing them as safe, caution, or danger [7]. The alerts include both short-and long-range communication and are conveyed via the LEDs, alarms, Bluetooth, and GSM. It's modular construction provides flexibility in deployment and integrates easily into an existing network. The prototype has been tested and demonstrated accurate monitoring and alerting, thus providing a foundation for flood disaster mitigation.

The blending of IoT with ultrasonic sensors, or rain sensors, makes it possible to mitigate flood occurrences through a real-time monitoring system and messaging information through SMS messages, even in remote areas [8]. While the system gives timely alerts, its dependency on GSM infrastructure as well as the high initial hardware costs made are some notable challenges that need to be addressed. Nonetheless, the same data is processed in real time with community engagement, which makes it improve disaster response.

Another innovative approach involves combining sensors like the LM35 Temperature Sensor, MQ2 Gas Sensor, and LDR with GSM technology for developing enhanced safety applications [9]. These enable alert systems that send their alarms almost instantly for a very fast response and may be integrated in smart homes. Ultrasonic sensors make the fire-fighting robots mobile devices; automatic smoke detection promotes safety against fire. On the other hand, apart from all these advantages of real-time notification, issues like false alarm and power dependency still prevail in it. This may be the first proof that IoT and sensors could change and revolutionize the detection of fires in the unmonitored space.

Wildfire detection systems have been analyzed, and there have been some challenges within existing solutions including high cost and inefficiency and scalability [10]. These challenges are tackled with the introduction of ”PyroVision”, a Bluetooth-based GPS-free grid-based system. This study aims to demonstrate its cost-effectiveness, scalability, and performance in the field-a demonstration of its possible real-world applications.

Another work talks about the varied methodologies for cyclone and flood detection, including Remote Sensing, IoT, ANN, YOLO, and Wireless Sensor Networks [11]. Remote Sensing gives accurate detection but costly, and IoT, with its cost advantage, suffers from its accuracy. ANN and YOLO are resource-hungry and are accurate methods whereby WSN provides all-time monitoring but very careful of network coverage. The design focus is of cost-effective real-time developments using low-cost sensors and mobile devices for innovations in panic alarms or cyclone warnings.

The research describes several flood detection techniques along with the merits and demerits of each [12]. UAV imaging has high resolution available, but access is limited because it has special equipment. The dynamic flooding captured by video analysis is computationally expensive. Computer vision through geo-tagged images gives fast data but is oriented on the problems of orientation. Based on adoption, satellite and video analysis are adopted, but the paper provides an alternative option for crowd sourced image with color-based segmentation that is more accessible and cost-effective for people.

The work related to the development of voice-controlled robotics finds a connection with those related to speech recognition, helping robots in navigating and avoiding obstacles in more than 100 languages [13]. A system is under development that supports 119 languages for aiding in cases of fire and smoke emergencies [14]. Sensor inaccuracies are still a challenge in obstacle detection [15]. Recent developments are based on voice commands combined with Bluetooth capabilities with real-time possibilities of obstacle detection [16]. Forward progress would improve upon these systems in terms of reliability and accuracy.

Forecasting is vital for addressing disasters. Disaster forecasting determines climate malfunctions, floods, earthquakes and avalanches [17]. IoT sensors situated in places at risk of these occurrences are used to gather information about their surroundings. This collected data is then used by algorithms to predict the disasters and advise on what immediate response action need to be taken. Indeed, forecasting advertises early warnings that appear to save people a mixture of time required for preparing [18]. Response duration reduces the scope of the disasters, minimizes their impacts, and uplifts the resilience and safety of the vulnerable region.

The robot can recognize and destroy dangerous gases in the atmosphere, including chlorofluorocarbons, carbon monoxide, and methane, to give post-disaster safety [19]. It identifies and reduces their concentration by taking the help of oxidizing agents such as oxygen and hydrogen peroxide using gas sensors [20]. This robot can move forward and look around hostile terrain like areas affected by flooding through Arduino UNO capacity [21]. Thus, it could provide an efficient and practical method to handle and care for poisonous gases after natural disasters [22].

III. Methodology

A. Component Selection

Arduino is used because of its real-time processing of tasks, better analog support to directly read signals from the sensors, and more power efficiency. HC-SR04 Ultrasonic sensor is ideal because of its accurate detection of the water level by using ultrasonic waves, Rain sensor is suitable to detect the presence of rain and its intensity. The MQ2 smoke sensor can detect smoke and fire for wide-ran gas and MQ4 methane gas sensor is highly sensitive to methane and other flammable gases which makes it useful for detecting gas leaks and preventing fire. HC-05 Bluetooth module is used for wireless communication that transmits messages to mobile devices via the Bluetooth terminal app. LEDs represent visual indications and buzzer for distinct sound patterns.

B. System Mechanism

The readings collected from each sensor that is continuously monitoring the environmental conditions are transferred to the Arduino UNO which can processed in real-time. The Arduino processes these readings to determine whether any of the readings are higher than preset limits including increasing water levels, high gasses, presence of smoke, or rainfall among others. On passing a threshold, the HC-05 Bluetooth module sends an alert message to the mobile device of the user via the Bluetooth Terminal application. At the same time, LEDs glow in certain configurations, while the buzzer plays various pitch patterns corresponding to the identified type of disaster, so users receive instant and comprehensible notifications.

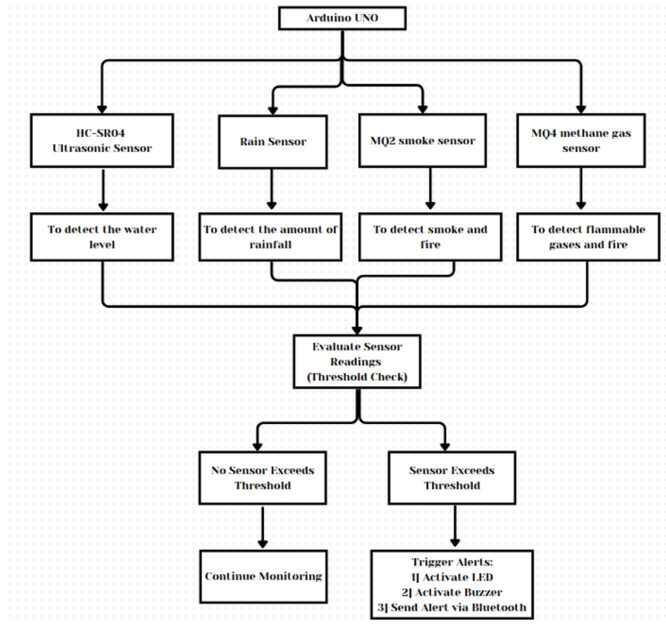


Fig. 1. Architecture of the Proposed Integrated Disaster detection System

Figure 1 is the block diagram that shows the usage of Arduino UNO along with various sensors and Bluetooth, LEDs, and Buzzer to detect floods and fire. The main controller which is the Arduino Uno collects data from various sensors present which are ultrasonic HC-SR04 and rain sensor for flood and rainfall detection. The other sensors namely the MQ2 smoke sensor and MQ4 methane gas sensor are used to detect smoke and fire. These sensors are well-connected with the Arduino in terms of power pin, ground and data pins to enable the quick identification and responding mechanism. The beauty of this design is that an Internet connection is not required in the process and it keeps the cost and operations of the system simple.

C. Hardware used

1) Arduino Uno - the central processing unit
2) HC-SR04 Ultrasonic Sensor - To detect water level
3) Rain Drop Sensor - To detect rainfall
4) MQ2 Smoke Sensor - To detect fire or smoke
5) MQ4 Methane Sensor - To detect gas leakage
6) HC-05 Bluetooth Module - To send alerts to the Bluetooth Terminal app

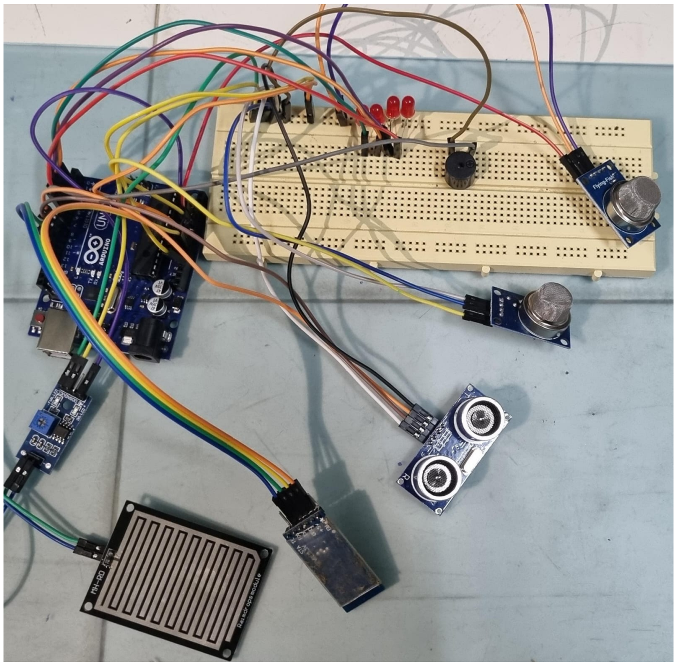


Fig. 2. Hardware Setup for the Proposed Integrated Disaster detection System

7) LEDs - For visual indication
8) Buzzer - For distinct audio alerts based on the type of disaster
9) Connecting wires

D. Software used

1) Arduino IDE - To code and upload program
2) Libraries to handle the Bluetooth communication using the HC-05 module

IV. Implementation

The section provides a detailed description of the hardware platform that was employed in the disaster detection and alerting system. The system connects various sensors and parts with the Arduino UNO to measure the environmental conditions and signal fires and flood danger. All the components have their own significant purpose of identifying the real-time hazard and communicating to the user.

Figure 2 illustrates the full hardware setup of the system. Arduino UNO communicates with all the sensors and the components and initiates the detection and alerting system. The HC-SR04 ultrasonic sensor determines a change in the water level by transmitting ultrasonic waves from the Trig pin connecting to digital pin 8 and receiving them on the Echo pin connecting to digital pin 9 which estimates the distance to the water surface. The Rain Sensor which is plugged into analog pin A2 converts the level of rainfall intensity to resistance by the change on the conductive path. The MQ2 Smoke Sensor and MQ4 Methane Sensor are connected to the analog pins A0 and A1 pins, respectively, where the smoke and flammable gases measure air quality by determining the smoke and methane gas concentration in the vicinity. The HC-05 Bluetooth Module connected to digital pins 10 and 11 allows for the transmission of alert messages to a mobile application once any of the hazards are identified. For the visual alerts, LEDs numbered 2, 3, 4, and 5 light up for water level rises, smoke, gas leaks, and rainfall respectively while for audible alerts there is a Buzzer on pin 6 that produces different beeps for the four events. All components employ a common ground linked to the Arduino Ground pin, with power being supplied through USB and concurrent real-time hazard identification in diverse surroundings.

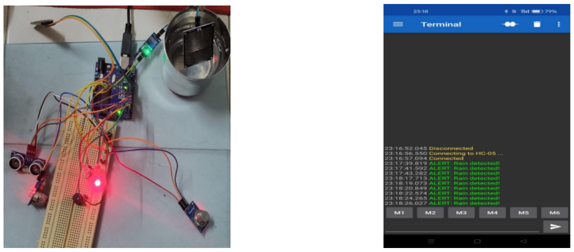
V. Results

The study effectively detects environmental threats such as high water levels, gas leakage, smoke, and rising rainwater levels through the incorporation of multiple sensors. The HC-SR04 Ultrasonic Sensor does a good job of reading the water level correctly and alerts the system whenever the distance to the water falls below the set limit. The rain sensor responds immediately to a certain rainwater level rise. The MQ2 and MQ4 sensors operate efficiently to identify high levels of smoke and gas, respectively, with the system providing corresponding notifications through the HC-05 Bluetooth module to a mobile application. There is a light and different sound warning (LED and buzzer) for each disaster detection.

A. Flood Detection

Figure 3 depicts the detection of flood through Rain Sensor. The Rain Sensor connected to the Arduino UNO will sense rainfall by measuring the water level through its analog output. The sensor's output voltage changes when the water level crosses a predefined threshold, and Arduino will read the data. If the water level exceeds the set limit which implies there is rain then the Arduino processes the signal and connects to the HC-05 Bluetooth module which transmits a ”Rain has been detected” message from the Arduino to the linked smartphone using the Bluetooth Terminal app. The system at the same time turns on specific LED, which blinks to indicate a detected condition and the Buzzer is programmed to have 4 pulses of varying beeps for this audible signal of an alarm. This sensor works with the fact that the resistance changes depending on how water touches the sensor. Once the water level exceeds a certain amount, the sensor will send an alert to the users, which will ensure that users are informed of flooding or heavy rain conditions.

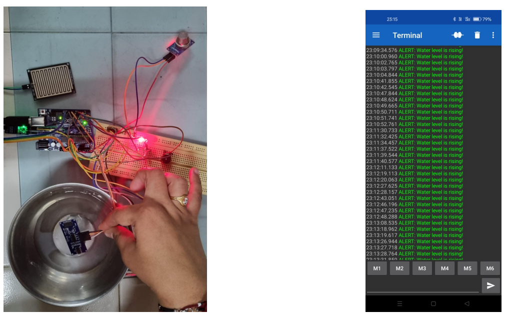
Figure 4 depicts the detection of flood through ultrasonic sensor.The HC-SR04 ultrasonic sensor issues an alert when the distance from or between the sensor and the water level is below a given threshold. The ultrasound waves from the sensor measure this distance and also depend upon the time taken for the echo to return. When the water level goes up near the proximity of the sensor, the system alerts a mobile device by sending a message ”Water level is rising!” through HC-05 Bluetooth module. It also flashes on an LED light, representing the impending danger, and starts a continuous beeping sound using the buzzer for easy attention. This entire usage,



3a) Detection through Rain Sensor   3b) Alert messages in the application

Fig. 3. Rain Sensor Detection

therefore, provides very quick rescue from flooding conditions, giving very justifiable alerts by sound and visibility.



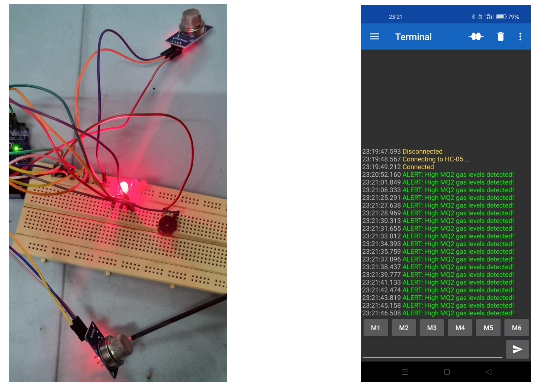
4a) Detection through Ultrasonic sensor   4b) Alert messages in the application

Fig. 4. Ultrasonic Sensor Detection

B. Fire Detection

Figure 5 depicts the detection of fire throuch MQ2 gas sensor. Connecting a MQ2 gas sensor with an Arduino UNO provides continuous air monitoring for smoke and combustible gases, measuring the resistance change relative to the gas concentration in the air. Once the MQ2 reading exceeds a set limit, it is processed by the Arduino UNO, which then causes the HC-05 Bluetooth module to send an alert message of ”High MQ2 gas levels detected” through the Bluetooth Terminal app. An LED turns on and the Buzzer will beep twice to alert the people around. The MQ2 gas sensors sense smoke or flammable gases with relative changes in the resistance of the preset threshold. Once the concentration exceeds this set level, there will be an abrupt output signal of MQ2 such that it will prompt an alert system through Arduino.

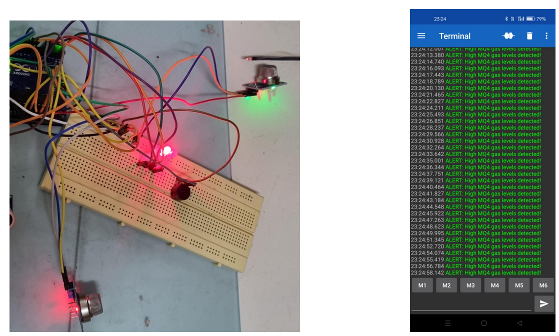
Figure 6 depicts the detection of fire throuch MQ4 gas sensor.The MQ4 Methane Gas Sensor connected to the Arduino UNO measures the concentration level of methane and other flammable gases in the surrounding air through changes in resistors, sensing material that varies differently depending on the gas exposures while an alarm is triggered the moment the concentration level in the air exceeds the threshold value. The HC-05 Bluetooth module sends a message saying ”High MQ4 gas levels detected” to the mobile device through the Bluetooth Terminal app, which notifies the user of the dangerous levels of gases. In parallel, the system will have specific LED with



5a) Detection through MQ2 Gas Sensor   5b) Alert messages in the application

Fig. 5. MQ2 Gas Sensor Detection

light on indicating a gas hazard and will have the Buzzer emitting two different beeps for an audible alarm. This sensor MQ4 functions by detecting gases present in the air and transmitting this through the Arduino. In this case, after gas concentration has surpassed a specific threshold, the system sets in alert mechanisms that comprise the flashing of LED lights, broadcasting on Bluetooth messaging, and creating buzzing noise from the buzzer.



6a) Detection through MQ4 Gas Sensor   6b) Alert messages in the application

Fig. 6. MQ4 Gas Sensor Detection

VI. Conclusion

The proposed work implements the HC-SR04 Ultrasonic Sensor, Rain Sensor, MQ2 Smoke Sensor, and MQ4 Methane Gas Sensor with the Arduino UNO, enabling a viable disaster detection and signaling measure for fire and flood risks. It functions by constantly collecting the physical input values like water level, rain, smoke, and gas concentration, and as soon as the input conditions meet set limits, the HC-05 Bluetooth Module immediately sends instant messages to a mobile phone via a Bluetooth terminal application and activates LEDs and the Buzzer are well. The positive aspects of the system are the high efficiency in identifying certain disasters and moderate price and the proposed system does not need the internet connection for the bluetooth wireless communication. Subsequent works can include the extension of enhancing the communication range with technology like Wi-Fi or LoRa in case of remote surveillance. In addition, refining the capability of sensors, incorporating machine learning algorithms for improving the thresholds, and programming control mechanisms. The system is cost-effective and scalable, which makes it suitable for rural as well as urban environments.

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